Faculty of Engineering and Technology

Agra College, Agra

CT1

Digital Image processing (MTCS 052)

M Tech

II Semester

MM: 20 Time: 1 hours

Note: Section A (Attempt any 2 questions, 2.5 marks each), Section B (Attempt any 1 questions, 5 marks each) and Section C (Attempt any 1, 10 marks).

Section A

Que1. What is digital image processing?

Que2. Explain computer graphics and computer vision.

Que3. What is sampling and quantization?

Section B

Que7. Explain types of distortion in image.

Que8. Explain Convolution operation with an example.

Section C

Que10. Perform histogram specification

Image gray level distribution

Gray level	0	1	2	3	4	5	6	7
No of pixel	8	10	10	2	12	16	4	2

Target Histogram

Gray level	0	1	2	3	4	5	6	7
No of pixel	0	0	0	0	20	20	16	8

Que. 11 Explain key steps of digital image processing.

Bahutio
vilnaale

Faculty of Engineering & Technology
Agra College, Agra
Class Test-I

Subject Name:- SPPM
Branch:- M.Tech.
Year/Sem. :- II Sem.

Subject Code :-
Duration :- 60 Minutes
Max. Marks: - 20

Note:
⇒ Attempt any three questions. All question carry equal marks.

Q1. Define Project. Explain its characteristics.

Q2. Explain Project Life Cycle Models.

Q3. Define project management and it's objective in detail.

Q4. Explain the roles and responsibilities of a Project Manager.

Q5. Describe different Project Life Cycle Models such as Waterfall, Spiral, and Agile.

Faculty of Engineering & Technology
Agra College, Agra
Class Test -1

Subject Name: Wireless and Mobile Networks
Branch : M.Tech CS
Year/Semester: 1st/2nd

Subject Code: MTCS-~~4??~~ 202
Duration: 1 hour
Maximum Marks: 20

Section – A

Note: Attempt all questions. (1.5X5=7.5)

Q1. Define multiplexing?

Q2. Define modulation?

Q3. What is GSM?

Q4. What does FDMA stand for?

Q5. What are wireless and mobile networks.

Section – B

Note: Attempt any two questions. (2.5X2=5)

Q6. Describe the concept of TDMA.

Q7. Explain the purpose of TETRA?

Q8. How does spread spectrum improve signal reliability?

Section – C

Note: Attempt any one questions. (7.5X1=7.5)

Q9. Compare the features and capabilities of LTE and UMTS in terms of data transmission and network architecture.

Q10. Discuss the advantages and disadvantages of CDMA compared to other medium access control techniques.

F.E.T. Agra College, Agra

M.Tech (CS-2nd semester CT-!st)

Session 2025-26.

Subject Name :Multicore Architecture & Prog.. **M.M. 20**

Subject Code:MTCS- 201 **M.T. 1 Hr.**

Note: Attempt any Four Questions

1. What are different Parallel Computing Plateforms . Explain
2. Differentiate betwwen Multicore Architecture and Hyper threading technology
3. Explain Multithreading on Single Core Versus Multicore plateforms .
4. What is Concurrency and how it motivates in software ,
5. Explain Amdals law and Gustafsons law.

P.T.O.

Faculty of Engineering & Technology, Agra College, Agra

2026

CLASS TEST # 2

Subject	: MACHINE LEARNING	M.M.	: 20 Marks
Subject Code	: MTCS-031	SEMESTER:	II ND
Course / Branch	: M.TECH. / Comp. Sc & Engg.	Time	: 1 Hour

NOTE:- Attempt ALL Sections. Question's within each Section carry equal marks.

SECTION - A | SHORT ANSWER TYPE QUESTIONS

Q(1) :Attempt ANY FIVE PARTS : (2 MarkS x 5)

(a) Differentiate b/w Supervised Learning & Unsupervised Learning ?

(b) Differentiate b/w Conventional Computers & Machine Learning Systems ?

(c) Differentiate between Clustering & Classification ?

(d) Write FIND-S Algorithm ?

(e) Write List Then Eliminate Algorithm ? awane

(f) Write Candidate Elimination Algorithm ?

SECTION - B | MODERATE SIZE QUESTIONS

Q(2) : Attempt ANY TWO PARTS : (3 Marks x 2)

(a) Explain Concept of FIND-S Algorithm with the help of suitable example and diagram ?

(b) Explain Concept of Candidate Elimination Algorithm with the help of suitable example and diagram ?

(c) Explain the following in brief with suitable examples ?

(i) Concept Learning
(ii) Inductive Learning Hypothesis
(iii) General to Specific Ordering of Hypothesis

(d) Differentiate between Single Layer ANN and Multi Layer ANN with the help of suitable diagrams ?

SECTION - C | LENGTHY QUESTIONS

Q(3) : Attempt ANY ONE PART : (4 Marks x 1)

(i) Write Perceptron Learning Algorithm in ANN ?

(ii) Explain Back Propagation Learning Algorithm in ANN ?

(iii) Explain Decision Tree Learning ?

(iv) Discuss Applications of Machine Learning in the various areas of science and technology ?

Faculty of Engineering & Technology, Agra College, Agra

2026

CLASS TEST # 1

Subject : MACHINE LEARNING
Subject Code : MTCS-031
Course / Branch : M.TECH. / Comp. Sc & Engg.

M.M. : 30 Marks
SEMESTER: II ND
Time : 1 Hour

NOTE:- Attempt ALL Sections. All Question's of each Section carry equal marks.

SECTION - A	SHORT ANSWER QUESTIONS

Q(1) :Attempt ALL PARTS : (2 MarkS x 5)

(a) Define Machine Learning in terms of Experience (E), some class of Task (T), and Performenance Measures (P) ?

(b) Give the Partial Design of Checkers Learning Program ?

(c) Differentiate b/w Supervised Learning & Unsupervised Learning ?

(d) Differentiate b/w Conventional Computers & Machine Learning Systems ?

(e) Differentiate between Clustering & Classification ?

SECTION - B	MODERATE SIZE QUESTIONS

Q(2) : Attempt ANY TWO PARTS : (5 Marks x 2)

(a) Define the term Task (T), Performenance Measures (P), and Training Experience (E) for the following Machine Learning Problems :

(i) A Checkers Learning Problem
(ii) A Handwriting Recognition Learning Problem
(iii) A Robot Driving Learning Problem

(b) Explain the following in brief with suitable examples ?

(i) Concept Learning
(ii) Inductive Learning Hypothesis
(iii) General to Specific Ordering of Hypothesis

(c) Differentiate between Artificial Neural N/W & Biological Neural N/W with the help of suitable diagrams ?

(d) Design ANN for following Boolean Function by using Mc-Culloch Pitts Neuron Model ?

$$Y = f(x_1, x_2, x_3) = x'_1 . x_2 . x'_3$$

SECTION - C	LENGTHY QUESTIONS

Q(3) : Attempt ANY ONE PART : (10 Marks x 1)

(a) Discuss the Final Design of Checkers Learning Program ? Also give Summary of choices in designing the Checkers Learning Program ?

(b) ATTEMPT ANY FOUR PARTS :

(i) Write K - Means Clustering Algorithm ?

(ii) Write K - Nearest Neighbour Algorithm ?

(iii) Discus any Five Major Issues in Machine Learning in nutshell ?

(iv) Differentiate between Single Layer & Multi Layer Feed Forward Neural N/W with the help of suitable diagrams ?

(v) Explain Hebb's Rule used in Neural N/W and write Hebbian Learning Algorithm ?